WEEK 4

Network Protocol Attacks Lab

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Date of Submission: 03.04.26

Network Protocol Attacks Lab Report

1. Aim

To perform a Man-in-the-Middle (MitM) attack in a controlled lab environment and analyze insecure network protocols by capturing sensitive data using packet analysis tools.

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2. Tools Used

- Ettercap (for ARP Spoofing)
- Wireshark (for packet analysis)
- Metasploitable 2 (target machine)
- Kali Linux (attacker machine)

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3. Lab Setup

The experiment was conducted in a host-only network using VirtualBox. Both Kali Linux and Metasploitable 2 were connected to the same network interface.

- Attacker IP (Kali Linux): 192.168.56.106
- Target IP (Metasploitable 2): 192.168.56.108

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4. Procedure

Step 1: Network Configuration

The IP addresses of both machines were identified using:

```
ip a
```

Step 2: Enable IP Forwarding

IP forwarding was enabled on Kali Linux to allow packet forwarding:

```
sudo sh -c "echo 1 > /proc/sys/net/ipv4/ip_forward"
```

Step 3: Launch Ettercap

Ettercap was started in graphical mode:

```
sudo ettercap -G
```

- Interface selected: eth1
- Unified sniffing enabled

Step 4: Host Discovery

Hosts were scanned using:

- Hosts → Scan for Hosts
- Hosts → Hosts List

The Metasploitable machine was identified.

Step 5: ARP Spoofing Attack

Targets were configured:

- Target 1: 192.168.56.108 (Metasploitable)
- Target 2: 192.168.56.106 (Kali)

ARP poisoning was started:

- Mitm → ARP poisoning
- Option: Sniff remote connections enabled

Step 6: Packet Capture using Wireshark

Wireshark was launched and the eth1 interface was selected.
Filter applied:

```
ftp
```

Step 7: Triggering Traffic

An FTP connection was initiated:

```
ftp 192.168.56.102
```

Login credentials used:

- Username: msfadmin
- Password: msfadmin

Step 8: Credential Capture

Using "Follow TCP Stream" in Wireshark, the following credentials were captured:

```
USER msfadmin
PASS msfadmin
```

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5. Results

Attack ID	Technique	Target IP	Status	Outcome
001	ARP Spoofing	192.168.56.108	Success	Traffic Intercepted
002	FTP Sniffing	192.168.56.108	Success	Credentials Captured

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6. Analysis

The FTP protocol transmits data in plaintext, making it vulnerable to interception. During the MitM attack, login credentials were successfully captured using Wireshark. This demonstrates the lack of encryption in FTP and highlights the risk of using insecure protocols.

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7. Conclusion

The lab successfully demonstrated a Man-in-the-Middle attack using ARP spoofing. Sensitive information such as FTP login credentials was intercepted, proving that unencrypted protocols are highly vulnerable. Secure alternatives such as SFTP or HTTPS should be used to prevent such attacks.

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8. Limitations

SMB relay and NTLM hash capture could not be performed because Metasploitable 2 is a Linux-based system and does not support Windows-based authentication protocols.

9. Learning Outcomes

- Understood the concept of Man-in-the-Middle attacks
- Learned ARP spoofing using Ettercap
- Analyzed network packets using Wireshark
- Identified vulnerabilities in FTP protocol
- Captured plaintext credentials
- Understood importance of encryption
- Gained hands-on experience in network security
- Learned packet filtering techniques
- Understood risks of insecure protocols
- Improved practical cybersecurity skills